GovConnect.lk

GovConnect.lk – AI & ML Powered Citizen Complaint Management System

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PROBLEM

Common problems include:

- • Complaints are manually sorted and assigned
- • Slow response time from authorities
- • Lack of transparency in complaint handling
- • Citizens cannot easily track complaint progress

Research Objective

- Automatically detect the correct government department for complaints
- Prioritize complaints using an intelligent algorithm
- Allow citizens to vote for important issues
- Enable government agents to manage and update complaints efficiently
- Provide transparency through dashboards and status tracking

PROPOSED SOLUTION

Key features include:

- Citizens can submit complaints online
- The system automatically predicts the relevant department
- Complaints are prioritized using machine learning
- Agents manage and resolve complaints
- Ministers monitor complaints through dashboards

SYSTEM WORKFLOW

- Citizen registers and logs into the system
- Citizen submits a complaint
- AI predicts the relevant department
- Complaint is posted to department dashboard
- Priority model calculates urgency score
- Complaint is assigned to an agent
- Agent updates complaint status
- Minister monitors complaints through dashboard

AI COMPONENTS

Two AI models are integrated in the system.

Department Prediction Model

- Uses RAG with Llama 3
- Analyzes complaint text
- Predicts the most relevant government department

Priority Prediction Model

Uses Random Forest Regression

Calculates urgency score using features such as:

- Votes
- Keywords in complaint title
- Recency (days)
- Complaint status

The urgency score is used to rank complaints.

TECHNOLOGY STACK

Frontend
- React.js (Vite)
- Tailwind CSS 4

Backend
- Node.js
- Express.js (MVC Architecture)

Database
- MongoDB

Authentication
- JWT Authentication
- Google OAuth

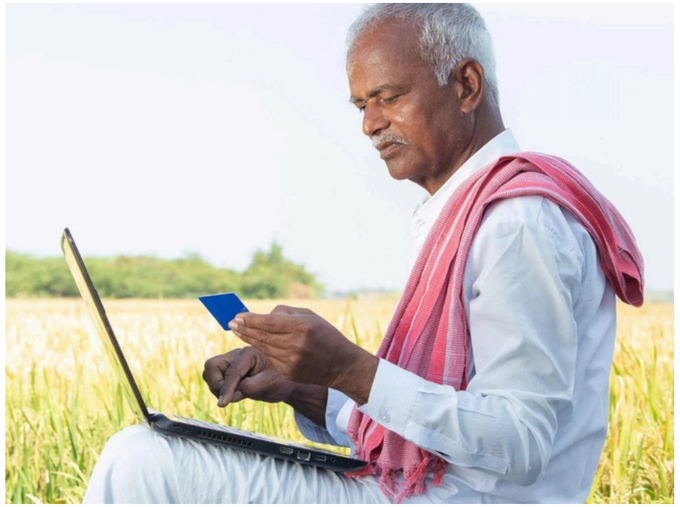
AI Components
- Random Forest Regression – complaint priority prediction
- RAG + Llama 3 – department prediction

SYSTEM WORKFLOW

System operation process:

1. Citizen logs into the system
2. Citizen submits a complaint
3. AI predicts the correct department
4. Complaint is posted to department dashboard
5. Priority model calculates urgency score
6. Complaint assigned to an agent
7. Agent updates complaint status (Pending, Ongoing, Resolved, Rejected)
8. Minister monitors complaints through dashboard

Target Group



- **Citizens submitting complaints through the Citizen Portal**
  Urban and rural populations, including those with low digital literacy

- Minister create a department and the assign the agents

- The agent resolved the issue within the resolution time

SOLUTION OVERVIEW

Component	Owner	**Feature**
Random Forest Priority Predictor	W.M.B.S Premalal	• Developed the Machine Learning Priority Prediction Model using Random Forest Regression to calculate complaint urgency scores. • Designed and implemented the Agent Logical Dashboard for managing assigned complaints and updating complaint status efficiently.
Department Predictor & Citizen Voting Module	D.M.C Shehhara	Implemented the RAG-based Department Prediction System using Llama 3 to automatically identify the relevant government department. • Developed the Citizen Voting System to allow users to vote for complaints, helping highlight high-priority public issues.

NON-FUNCTIONAL FOCUS

- **Performance**: Handle 10,000+ users, <3 sec response time for complaints, <2 sec processing for classification & prioritizaion
- **Scalability**: Scale to 50,000+ complaints/month
- **Reliability:** 99.9% uptime, resilient to failures with failover mechanisms.
- **Availability:** 24/7 access, especially for rural areas & emergencies.
- **Usability:** Simple interface, multilingual support (Sinhala/English), real-time complaint tracking.
- **Security:** Data encryption, role-based access control, GDPR-like privacy compliance.
- **Maintainability:** Modular system, clear API documentation, easy updates.
- **Interoperability:** Integration with government systems, API-based data exchange.

Commercialization/Sustainability of Solution

GovConnect LK is a digital governance solution for local councils and ministries that automates complaint handling, reduces manual work, lowers operational costs, and offers a scalable, sustainable platform for long-term nationwide e-governance.

Customer Persona

- Citizens: Urban & rural users submitting complaints (Sinhala–English)
- Government Officers: Handling, prioritizing, and resolving complaints
- Administrators: Managing councils, policies, and analytics

Market Size

- All local authorities and ministries in Sri Lanka
- Expandable to provincial and national e-government systems
- Applicable to other developing countries with similar needs

What is Unique in the Solution

- Uses Agentic AI instead of a single AI model
- Supports Sinhala–English code-mixed complaints
- Hybrid prioritization using:
  - **Urgency**
  - **Predicted resolution time**
  - **Community voting**
- Transparent, auditable decision-making

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THANK YOU!

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